WORKED SOLUTION

STEP II 1992, Question 16

Probability and Statistics • Conditional Probability

STEP II 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.17, *Banach's Matchboxes*: https://beyondstep.keithmcnulty.org/ch12/matchboxes.html

The key idea

This is a version of Banach's matchbox problem. Since we're told *when* the shortage was discovered (on reach N), this is a conditional probability. So we compute two joint probabilities: that the first failure happens on reach N and is a toffee, and that it happens on reach N and is a mint. Each one is a "first $N-1$ reaches contain exactly so many of each kind, then one particular kind" calculation, which is binomial.

Solution

Assume the reaches are independent, each a toffee with probability p and a mint with probability q, where $0 < p < 1$. Let T be the event that on the Nth reach she reaches for a toffee and finds the toffee packet empty, and that she has found no packet empty on an earlier reach. Let M be the same event with mints in place of toffees. We want

$$\mathrm{P}(\text{toffee} \mid \text{first discovery on reach } N) = \frac{\mathrm{P}(T)}{\mathrm{P}(T) + \mathrm{P}(M)}.$$

The probability of T

For T to happen, the toffee packet must be empty after $N-1$ reaches. No reach so far has failed, so every one of those reaches took a sweet. So exactly n of the first $N-1$ reaches were for toffees (one more would have been a failure), and the other $N-1-n$ were for mints. Then reach N is for a toffee.

Conversely, suppose the first $N-1$ reaches contain exactly n toffee-reaches and $N-1-n \leq m$ mint-reaches, in any order. Then no earlier reach can have failed: at no stage had she reached for more than n toffees or more than m mints. Then if reach N is a

toffee, T happens.

Ask yourself Doesn't the order of the first $N-1$ reaches matter, so that she doesn't run out early?

No. She only *discovers* that a packet is empty when she reaches into it while it is empty. Emptying the toffee packet early (on the nth toffee) is fine as long as she doesn't reach for a toffee again before reach N. That is automatic, because the first $N-1$ reaches contain only n toffee-reaches. So all $\binom{N-1}{n}$ orders are allowed, and

$$\mathrm{P}(T) = \binom{N-1}{n} p^n q^{N-1-n} \cdot p = \binom{N-1}{n} p^{n+1} q^{N-1-n}, \qquad n \le N-1 \le n+m,$$

and $\mathrm{P}(T) = 0$ otherwise. If $N-1 < n$, she can't have used up the toffees. If $N-1 > n+m$, she would have needed more than m mints.

The probability of M

In exactly the same way, with the roles swapped,

$$\mathrm{P}(M) = \binom{N-1}{m} q^{m+1} p^{N-1-m}, \qquad m \le N-1 \le n+m,$$

and $\mathrm{P}(M) = 0$ otherwise.

The answer

The discovery must happen somewhere in reaches $\min(n,m)+1$ to $n+m+1$. Reach $n+m+2$ is impossible, because by then both packets would be empty and something would already have failed.

- If $N \le m$ (so $\mathrm{P}(M) = 0$ but $\mathrm{P}(T) > 0$), she was certainly reaching for a toffee: the probability is 1.
- If $N \le n$ (so $\mathrm{P}(T) = 0$), the probability is 0.
- If $\max(n,m)+1 \le N \le n+m+1$, both are possible, and

$$\mathrm{P}(\text{toffee}) = \frac{\binom{N-1}{n} p^{n+1} q^{N-1-n}}{\binom{N-1}{n} p^{n+1} q^{N-1-n} + \binom{N-1}{m} q^{m+1} p^{N-1-m}}.$$

In the last case we can tidy this up by dividing the top and bottom by $\mathrm{P}(T)$. The common factor $(N-1)!$ cancels from the binomial coefficients, and the powers combine to $(q/p)^{n+m+2-N}$. This gives

$$\mathrm{P}(\text{toffee}) = \left[1 + \frac{n!\,(N-1-n)!}{m!\,(N-1-m)!} \left(\frac{q}{p}\right)^{n+m+2-N}\right]^{-1}.$$

Checks

- Symmetry: if $n = m$ and $p = q = \frac{1}{2}$, the two expressions are equal and the answer is $\frac{1}{2}$, as it should be.
- Take $n = 2$, $m = 3$, $p = \frac{1}{2}$, $N = 5$. Then $\mathrm{P}(T) = \binom{4}{2}/2^5 = \frac{3}{16}$ and $\mathrm{P}(M) = \binom{4}{3}/2^5 = \frac{1}{8}$, so the answer is $\frac{3/16}{5/16} = \frac{3}{5}$. With fewer toffees, running out of toffees is more likely.
- $\sum_N \big(\mathrm{P}(T) + \mathrm{P}(M)\big) = 1$ over $N = 1, \ldots, n + m + 1$, because some reach must eventually fail. We checked this, and the formula, against an exact step-by-step enumeration for all $n, m \leq 6$ and several values of p. A 2×10^5-trial simulation with $n = 4$, $m = 3$, $N = 7$, $p = 0.55$ gave 0.527, against the exact 0.528.

Common slips

- Forgetting the final factor p (or q) for the Nth reach itself.
- Answering $\mathrm{P}(T)$ rather than the conditional probability $\mathrm{P}(T)/(\mathrm{P}(T) + \mathrm{P}(M))$. We are told that the discovery happened on reach N.
- Thinking the event is "the toffees run out on reach N". Taking the last toffee is not a discovery. The discovery is the next attempt.

Beyond STEP

The time of the first failure is a mixture of two *negative binomial* distributions. The negative binomial distribution counts the trials needed to get a given number of successes. Banach's original matchbox problem ($n = m$, $p = \frac{1}{2}$) asks how many matches are left in the other box at the moment of discovery. Its answer can be approximated with Stirling's formula, giving a mean of about $2\sqrt{n/\pi}$. For the negative binomial distribution, see Grimmett and Welsh, *Probability: An Introduction*. Feller's classic *An Introduction to Probability Theory and Its Applications*, Volume 1, has a famous discussion of the matchbox problem.

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